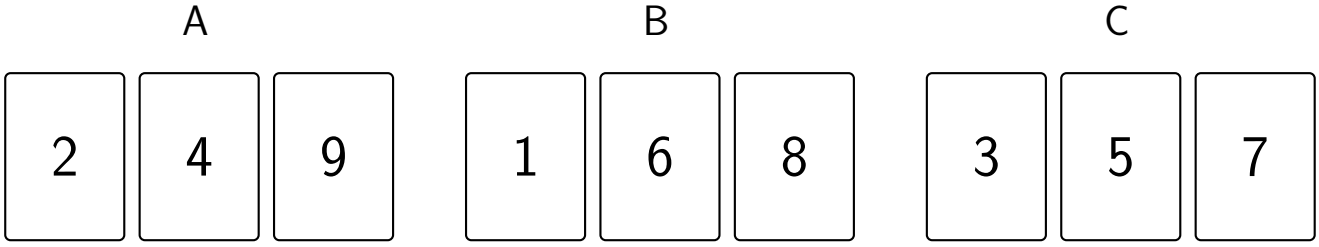
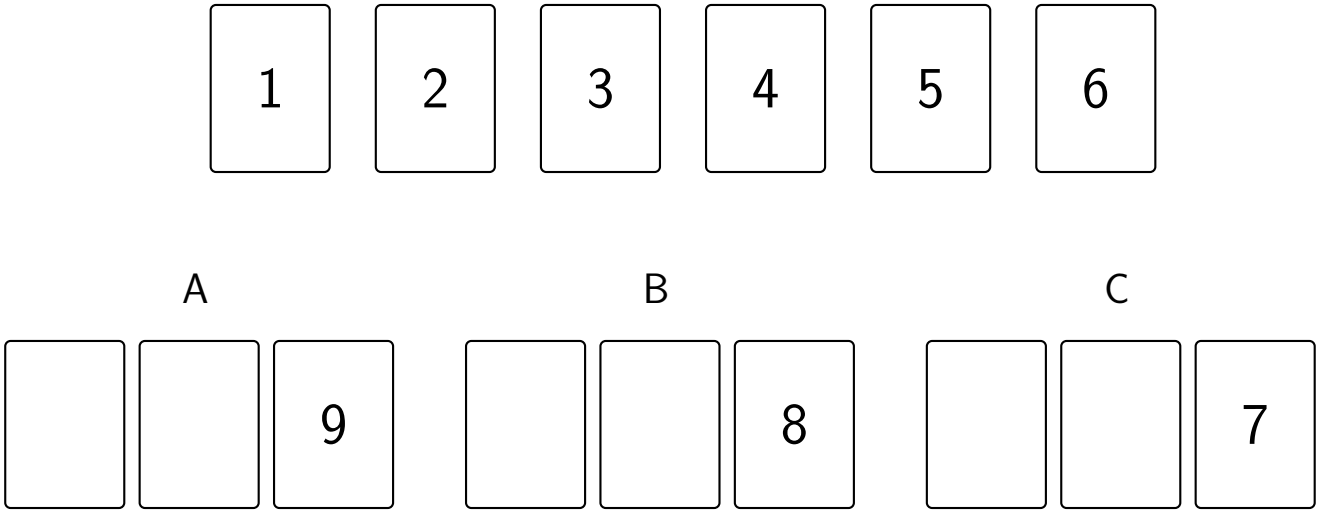
Each physical card is equally likely. Mix each deck separately and draw one card from each. Return both cards before the next round. The larger number wins. The two decks in a comparison must not share a number. Card order in a deck does not matter.

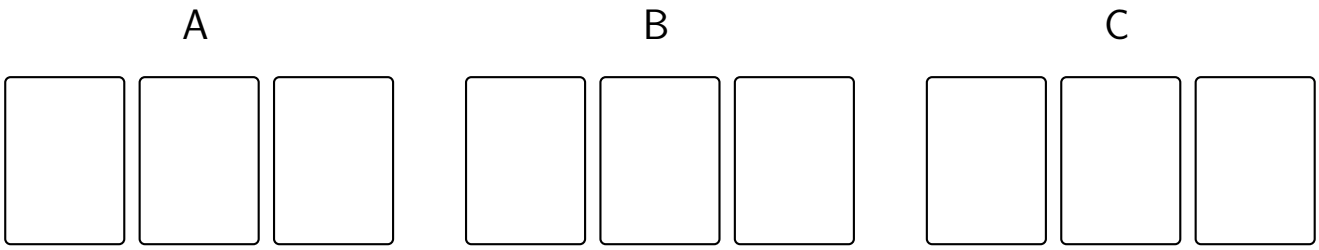
Problem 1: For each pair of decks, find how many of the possible card pairs each deck wins. Is there a deck that wins more than half its pairs against both others? Do the three deck totals settle this question?



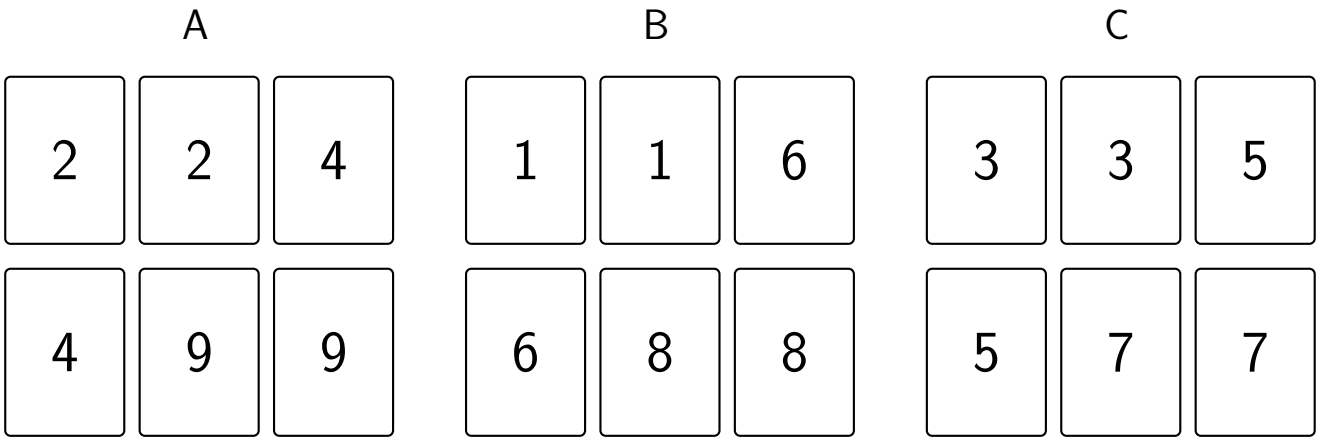
Problem 2: Keep 9 in A, 8 in B, and 7 in C. Use cards 1 to 6 once each to finish the decks so A wins more than half its pairs against B, B against C, and C against A. Find every arrangement and explain why none are missing.

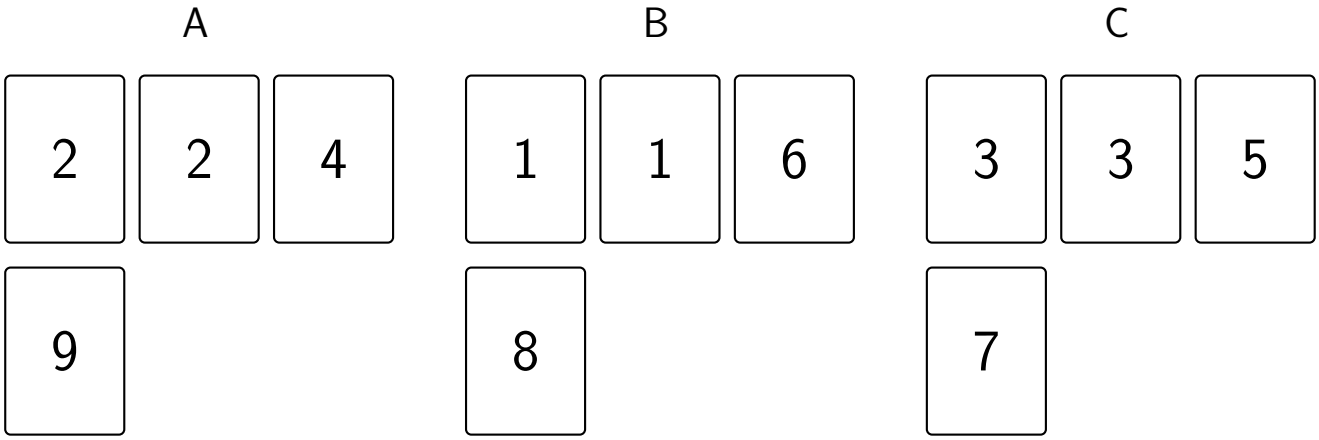


Problem 3: Write numbers from 1 to 6 on these nine cards. A number may repeat within one deck, but it cannot appear in two different decks. Make A win more than half its pairs against B, B against C, and C against A.

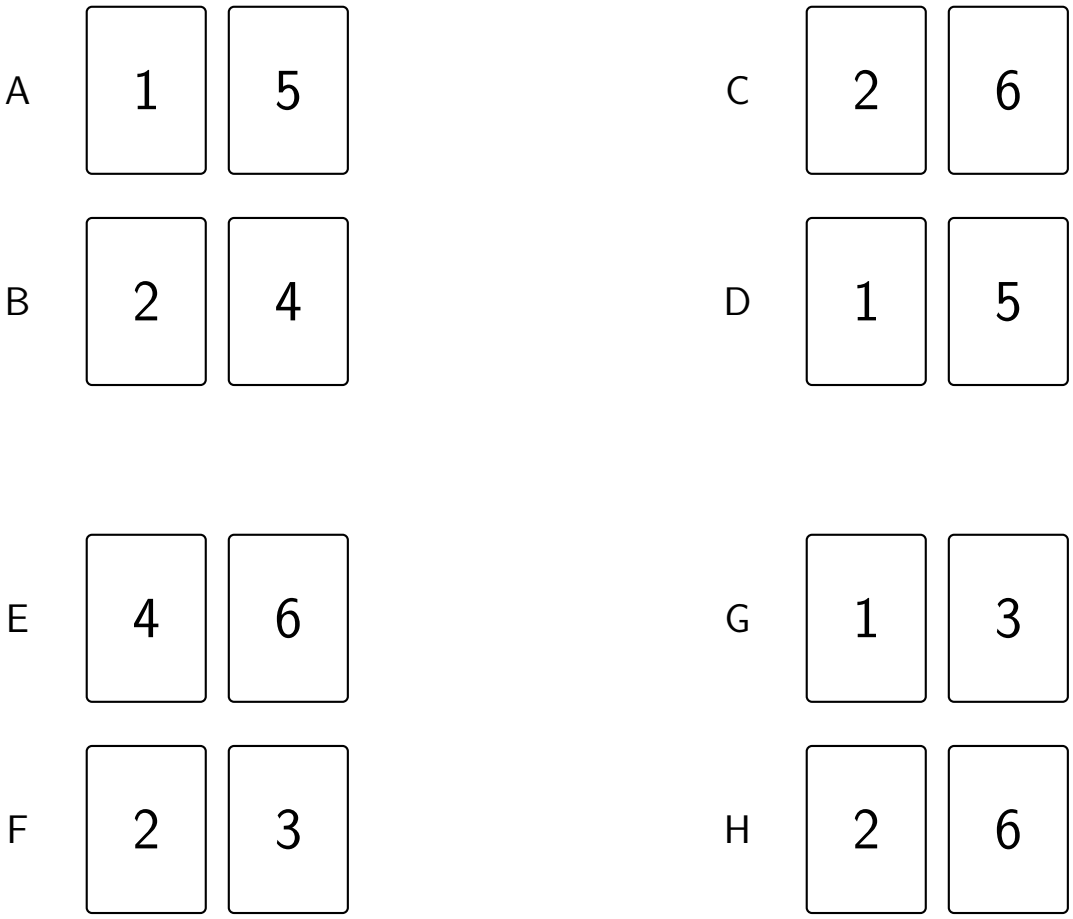


Problem 4: Find each deck's chance of winning against each other deck, first with the six-card decks and then with the four-card decks. Which chances changed from Problem 1?

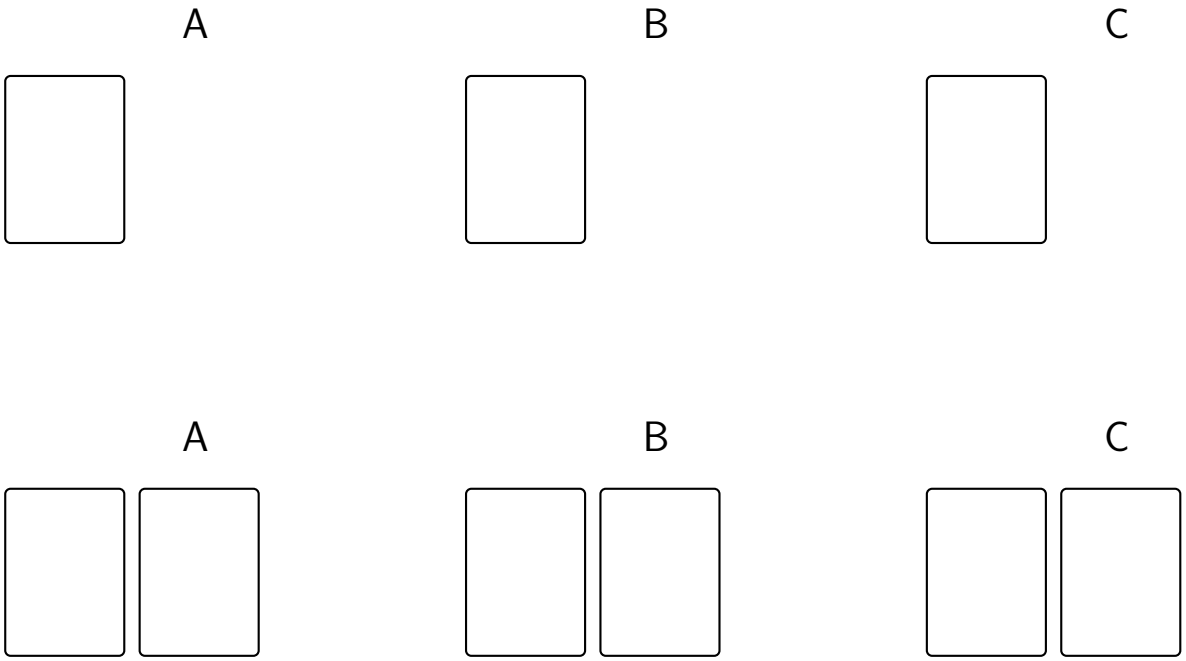




Problem 5: Decide which deck, if either, wins more than half the pairs in each comparison. Find a rule that works for every pair of two-card decks with no number shared between decks, and explain why it works.



Problem 6: Can three one-card decks make A win more than half its pairs against B, B against C, and C against A? Can three two-card decks do it? Explain for any choice of numbers. What is the fewest cards per deck that allows this, with all decks the same size?



Problem 7: Use cards 1 to 9 once each in three decks of three. Can A win at least six of its nine pairs against B, B at least six against C, and C at least six against A? Find an arrangement, or explain why it is impossible.

